

Human Behaviour as a Security Boundary: Affective AI and Behavioural Analytics for Cybersecurity

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Abstract—Small and Medium-sized Enterprises (SMEs) face a critical security paradox: they increasingly rely on digital supply chains yet lack the organisational readiness to maintain robust cyber defences. In the absence of dedicated security teams, informal trust-based communication becomes a primary vulnerability, making human behaviour the dominant attack surface. This paper proposes a behaviour-aware adaptive security architecture designed specifically for the resource constraints of SMEs. As a foundational exploration step, we treat human behaviour as a dynamic security boundary and evaluate our framework using emulated environments with synthetic agents and public datasets.

We present a modular architecture integrating: (1) Adaptive Behavioural Drift Detection that distinguishes between benign role changes and potential compromise without improved manual tuning; (2) Multi-Modal Fusion that correlates network, endpoint, and biometric signals to detect insider threats; and (3) Affective-Aware Interventions that tailor simulated security training to the user’s emotional state. We explore this architecture through four controlled experiments in a simulated SME setting, combining synthetic data and the CERT r4.2 benchmark dataset. In our insider-threat study on CERT r4.2, multimodal fusion improves detection Area Under the Curve (AUC) from 0.72 to 0.84 over a network-only baseline, while synthetic phishing simulations show that affective-aware training policies can approximately halve simulated click-through rates compared to a control condition. Finally, we discuss the architectural trade-offs between privacy, utility, and deployment complexity, offering a blueprint for future socio-technical security deployments in financially constrained environments.

Index Terms—human-centric cybersecurity, behaviour-aware security systems, multi-modal behavioural fusion, affective computing.

I. INTRODUCTION

Small and Medium-sized Enterprises (SMEs) comprise approximately 90% of the total number of businesses and 50% of the total number of employees globally [1]. SMEs are often affected by cyber attacks. A ransomware attack alone can cause the bankruptcy of a small-scale enterprise within six months due to the lack of financial flexibility, which larger-scale entities possess [2]. Nevertheless, the use of enterprise-grade security solutions in SMEs remains low. This is not just a technology issue but also a matter of organisational readiness. The inflexibility of security controls conflicts with the informality and trust that define the agility of SMEs. When security becomes a speed killer, employees often end

up disabling it altogether, creating a “Shadow IT” threat that defeats the purpose of security controls.

A. The Socio-Technical Gap

According to the Technology-Organisation-Environment (TOE) model, the impediment facing SMEs is not simply the availability of software, but organisational readiness. SMEs cannot easily manage “set and forget” tools or complex threat intelligence. What SMEs require is a security architecture that acts as a decision support system, translating requirements (e.g., GDPR, NIS2) into behaviour-centric mechanisms that accommodate rather than impede human activity. As an initial step toward building such a system, we propose evaluating a conceptual architecture within a controlled simulated setting.

B. Contribution

The paper bridges the “resource capability gap” by designing a *Human-Centric Security Architecture* for SMEs. Contrary to conventional security architectures that treat users as liabilities, we focus on human behaviour as a dynamic sensing and defence mechanism. Our specific contributions are:

- *Adaptive Behavioural Drift Detection*: We prototype and evaluate a stream clustering technique, Density-based Clustering (DenMune), for adaptive behavioural drift detection in synthetic SME activity logs, to differentiate between legitimate role evolution and malicious attacks without parameter retuning, attaining a 65% detection rate with a low false positive rate (Exp 1).
- *Multi-modal Insider Threat Fusion*: We demonstrate that fusing psychometric attributes with network and endpoint logs improves offline detection accuracy (AUC 0.84) compared to single-modality baselines (AUC 0.72) using the CERT r4.2 public dataset (Exp 2).
- *Affective-Aware Training Loop*: Through simulation, we empirically explore how personalising security training based on simulated emotional states can reduce synthetic agents’ susceptibility to phishing attacks compared to static training approaches (Exp 4).

C. Research Aim and Questions

The overall aim of this study is to develop the ground research for an AI-based cybersecurity system that empowers

SMEs to act as their own firewall by integrating behavioural intelligence. Acknowledging that real-world pilot implementation studies represent future work, this paper focuses on the technical feasibility via simulation, addressing the following research questions:

- **RQ1:** How can SMEs be supported to make trusted and resource-feasible cybersecurity decisions that incorporate human behaviour and organisational processes?
- **RQ2:** To what extent does the multi-modal fusion of psychometric traits with technical logs improve the offline detection of insider threats compared to single-modality baselines?
- **RQ3:** In synthetic phishing simulations, to what extent do affective-aware training policies influence simulated phishing susceptibility compared to cognitive-only or control policies?

Aspects related to real SME adoption, organisational readiness, and sustained practice are discussed conceptually but are not empirically evaluated in this paper; these remain open questions for future work, to be addressed through pilot deployments of the proposed architecture in SME environments.

II. RELATED WORK

A. Cybersecurity as a Socio-Technical Challenge

Cybersecurity in SMEs is a socio-technical challenge where the interplay between technology, humans, and processes is a key consideration [3]. Protection Motivation Theory (PMT) [4] and the Fogg Behaviour Model [5] propose a relationship between the perceived severity of threats and the "response cost" (the effort required to take action). In SMEs, where bandwidth is scarce, the high "response cost" of complex security tools leads to rational rejection [6].

Although these theories explain the reasons behind the lack of SME adoption, no technical framework has been proposed to integrate these factors into the design. Our work extends this by using "perceived effort" (friction logs) and "capability" (skill tracking) as parameters in the security engine.

B. User and Entity Behaviour Analytics (UEBA)

Traditional UEBA systems rely on baselines to identify anomalies [7]. Recent advances in deep learning (LSTM, Autoencoders) achieve high accuracy in log analysis [8].

Current UEBA literature focuses on maximising accuracy in a fixed dataset, ignoring the reality of "concept drift" in a growing SME. A system that flags every role change as a threat will create unmanageable levels of alert fatigue for a non-existent SOC team. In this work, we prioritise "drift adaptation" over sensitivity (Exp 1).

C. Affective Computing and Nudging

Affective computing [9] and nudging [10] are used in education and healthcare to drive long-term behavioural change. In cybersecurity, "fear appeals" are a common technique but are generally counterproductive, causing analysis paralysis rather than action [11].

In synthesis, current literature treats training and detection as separate silos. In reality, training and detection should be integrated. A user's emotional response to a training simulation should be correlated to their risk score, which in turn should be correlated to the nudges applied to them.

III. METHODOLOGY

A. Architectural Design Considerations

Designing a "Human Firewall" architecture requires balancing detection with strict resource constraints. The framework incorporates three key principles:

- 1) **Privacy-First Local Processing:** To mitigate surveillance concerns, raw behavioural data (e.g., keystroke dynamics) is processed locally on the endpoint. Only anonymised feature vectors or model updates are transmitted to the central engine.
- 2) **Modular Modality Ingestion:** The system algorithmically supports plug-and-play data sources. An SME can start with simple Network Logs and progressively add Biometrics as organisational readiness matures.
- 3) **Feedback-Driven Adaptation:** Behavioural baselines are dynamically updated based on the outcomes of training interventions, creating a conceptual closed-loop security system (see Fig. 1).

To clarify the boundaries of the current study, the privacy-first local endpoint processing and the real-time closed-loop deployment remain conceptual architectural goals designed for future real-world integration. In contrast, the underlying analytical engines—including the adaptive drift detection, multi-modal fusion network, and affective recognition algorithms—have been fully implemented and evaluated offline in this paper using simulated data and the CERT r4.2 dataset.

TABLE I
ARCHITECTURAL TRADE-OFFS FOR SME DEPLOYMENT

Decision	Benefit	Risk/Constraint
<i>Cloud vs. Edge</i>	Edge ensures privacy	Limited compute for Deep Learning models
<i>Active vs. Passive</i>	Active interventions train users	High friction causes fatigue
<i>Synthetic vs. Real</i>	Synthetic data allows rapid, ethical prototyping	Sim-to-Real gap in human entropy

B. System Architecture and Design Rationale

This research proposes a multi-layered cybersecurity framework designed for SMEs and incorporates behavioural analytics, multi-modal fusion, and affective computing. The proposed methodology is composed of four primary components that address different aspects of the human-centric security issue: adaptive behavioural drift detection, multi-modal insider threat detection, affective state detection, and affective-aware security training.

1) *Design Rationale:* We have chosen to use Density-based Clustering (DenMune) instead of sequence-based models (e.g., LSTM) for drift detection due to the absence of sufficient historical data in SME environments to train deep supervised models. We have chosen to use Feature-Level Fusion instead of Decision-Level Voting for the insider threat detection engine to ensure non-linear relationships between psychometric traits and network logs are effectively captured. Finally, given ethical and practical constraints, we extensively employ synthetic data emulation to validate the algorithms against worst-case scenarios, such as coordinated attacks.

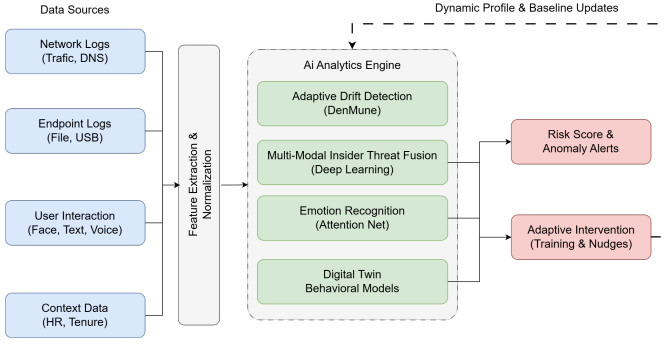


Fig. 1. Architectural Overview of the Human-Centric Cybersecurity Framework. Data flows from heterogeneous sources (left) into an AI Analytics Engine that performs drift detection, multi-modal fusion, and emotion recognition. The system outputs adaptive interventions and risk alerts (right) to create a dynamic “Human Firewall.”

C. Adaptive Behavioural Drift Detection

Traditional UEBA systems have difficulties with concept drift, where legitimate changes in user behaviour (e.g., new project, role changes) cause false positives. To address this, we have evaluated four different clustering algorithms: HDBSCAN, BIRCH, CluStream, and DenMune, in terms of their ability to detect drift from sudden anomalies in an online or semi-online manner.

We have formulated the drift detection problem as a stream clustering problem where user activities (login time, file access frequency, data transfer volume from external sources) are projected onto a high-dimensional feature space. A “drift” event is detected by identifying a user’s centroid that moves significantly from their historical centroid and remains in a new cluster for an extended period of time greater than a predetermined threshold (e.g., ≥ 24 hours).

In Algorithm 1, the decay parameter λ (set to 2.0) ensures that temporary anomalies cancel each other out twice as fast as they are generated, thereby preventing false alarms from accumulating short-term, unrelated anomalies.

D. Multi-Modal Insider Threat Detection

To detect sophisticated insider threats that may evade single-modality detection (e.g., network-only inspection), we proposed a Deep Learning Fusion architecture. The system integrates heterogeneous data sources:

- Network Logs: HTTP request volume.

Algorithm 1 Adaptive Behavioural Drift Detection

Require: User Stream $S = \{x_1, x_2, \dots, x_t\}$, Thresholds ϵ , T_{drift}

- 1: Initialize User Profile $P_u \leftarrow \text{Cluster}(S_{init})$
- 2: $State_u \leftarrow \text{NORMAL}$
- 3: $t_{abnormal} \leftarrow 0$
- 4: **for** each new event x_t in S **do**
- 5: $C_{curr} \leftarrow \text{AssignCluster}(x_t)$
- 6: **if** $\text{dist}(C_{curr}, P_u) > \epsilon$ **then**
- 7: $t_{abnormal} \leftarrow t_{abnormal} + \Delta t$
- 8: **if** $t_{abnormal} > T_{drift}$ **then**
- 9: $P_u \leftarrow C_{curr}$ {Update Baseline}
- 10: $State_u \leftarrow \text{DRIFT_DETECTED}$
- 11: **Trigger:** Adaptive Re-Authentication
- 12: $t_{abnormal} \leftarrow 0$
- 13: **else**
- 14: $State_u \leftarrow \text{WARNING}$
- 15: **end if**
- 16: **else**
- 17: $t_{abnormal} \leftarrow \max(0, t_{abnormal} - \lambda \cdot \Delta t)$ {Linear Decay}
- 18: $State_u \leftarrow \text{NORMAL}$
- 19: **end if**
- 20: **end for**

- Endpoint Logs: File modifications.
- Communication Data: Simulated email sentiment.
- Contextual Data Human Resources (HR): Psychometric scores (e.g., Big Five personality traits).

The model uses individual encoder networks for each modality to extract the features for each modality and then uses a fusion layer to classify the risk state of the user. This helps to capture complex non-linear correlations like a disgruntled employee (HR) exfiltrating data (Endpoint) during off-hours (Network).

E. Multimodal Emotion Recognition

Recognising affective states forms the basis of adaptive interactions. We implemented an offline Attention-based Fusion Network predicting emotions (e.g., Frustrated, Confused, Engaged) from simulated sensor outputs (facial embeddings, keystroke dynamic proxies, voice prosody). The attention mechanism weights the most informative modality dynamically.

- Facial Features: 128-D embeddings representing facial expressions.
- Textual Features: Sentiment scores and typing uncertainty from chat interactions.
- Keystroke Dynamics: Typing speed, pause rates, and backspace usage.
- Voice Prosody: Pitch variance and energy levels.

The attention mechanism learns to selectively focus on the most informative modality (e.g., voice pitch or keystrokes while silently working).

Algorithm 2 Multi-Modal Attention Fusion

Require: Modality Inputs $\{M_1, M_2, \dots, M_K\}$ (e.g., Face, Text)

```
1:  $H \leftarrow \emptyset$ 
2: for  $k = 1$  to  $K$  do
3:    $h_k \leftarrow \text{Encoder}_k(M_k)$  {Feature Extraction}
4:    $H$  append  $h_k$ 
5: end for
6:  $E \leftarrow \text{Concat}(H)$ 
7:  $\alpha \leftarrow \text{Softmax}(W_a \cdot \tanh(W_e \cdot E + b_e) + b_a)$  {Attention Weights}
8:  $V_{fused} \leftarrow \sum_{k=1}^K \alpha_k \cdot h_k$  {Weighted Sum}
9:  $y_{pred} \leftarrow \text{Sigmoid}(W_{out} \cdot V_{fused} + b_{out})$ 
10: return  $y_{pred}$ 
```

F. Affective-Aware Phishing Resistance Training

The final module evaluates—within an emulated environment—the premise of affective-aware training. Four conditions were modelled:

- 1) Control: Standard security awareness training.
- 2) Cognitive-Only: Focused on technical indicators of phishing (e.g., URL inspection).
- 3) Affective-Aware: Focused on recognising emotional triggers (e.g., urgency, fear) and impulse control.
- 4) Combined: Integrating both cognitive and affective modules.

The objective is to test the effectiveness of training users to identify their emotional response to the phishing lure as a countermeasure to reduce susceptibility to phishing attacks.

IV. EXPERIMENTAL EVALUATION

We designed four experiments evaluating our architecture on synthetic and public offline datasets.

- Exp 1 (Drift Detection): Addresses RQ1 via synthetic offline evaluation comparing stream clustering algorithms.
H1: User behaviour in SMEs exhibits "concept drift" that can be distinguished from malicious anomalies using density-based clustering (DenMune) without manual retraining.
- Exp 2 (Insider Threat Fusion): Addresses RQ1 and RQ2 by evaluating performance gains on the CERT r4.2 public dataset.

H2: Integrating psychometric traits (e.g., Neuroticism) with technical logs improves insider threat detection (AUC) compared to network-only baselines.

- Exp 3 & 4 (Affective Adaptation): Addresses RQ3 by measuring phishing susceptibility in a multi-agent simulation scenario.

H3: Security training interventions triggered by real-time affective states (e.g., stress) result in lower phishing susceptibility than static cognitive training.

A. Threats to Validity

Construct Validity: Using "Neuroticism" as a risk proxy in Exp 2 aligns with the CERT r4.2 ground truth but may oversimplify real-world psychological nuance.

External Validity (Synthetic Data Reliance): This work constitutes an architectural emulation. Experiments 1, 3, and 4 use synthetic user agents, not real humans. While we incorporated Gaussian noise ($\sigma = 0.15$) to mimic sensor degradation, simulated data inherently underestimates human entropy. Consequently, our findings support algorithmic feasibility, with real-world managerial efficacy heavily depending on future pilot SME validation.

B. Setup and Datasets

1) *Exp 1: Adaptive Drift Detection:* We generated a dataset consisting of 200 synthetic user agents spanning 60 days, modelling login and file access distributions.

- Control Group (40%): Stable behaviour profiles.
- Gradual Drift (30%): Profiles shifting incrementally over 14 days.
- Sudden Change (30%): Abrupt access shifts simulating promotion/demotion.

Algorithms (HDBSCAN, BIRCH, CluStream, DenMune) were compared using MTDD (Mean Time to Detect) and FPR (False Positive Rate).

2) *Exp 2: Multi-Modal Insider Threat on CERT r4.2:* We have used the CERT Insider Threat Test Dataset r4.2 [12], a well-known benchmark dataset that simulates 500 users in a corporate scenario over 18 months. Unlike synthetic data, CERT r4.2 contains:

- Psychometric Profiles: "Big Five" personality traits (Openness, Conscientiousness, Extroversion, Agreeableness, Neuroticism) for each user.
- Heterogeneous Logs: Logon, Device, File, HTTP, and Email activity.
- Ground Truth Scenarios: Detailed insider threat instances (e.g., sabotage by disgruntled IT admin).

This dataset is particularly interesting with regard to our "Human Behaviour" thesis since we are able to relate personality traits (e.g., high Neuroticism) to behavioural anomalies.

- Baselines: Single-modality XGBoost (Network-only).
- Fusion Model: We integrated psychometric scores ('O, C, E, A, N') as the "HR/Context" modality in our fusion network.
- Target Metric: Area Under the Curve (AUC) and F1-Score.

3) *Exp 3: Multimodal Emotion Recognition:* We simulated 150 learners participating in 8 security training sessions. Sensor data (Face, Text, Keystroke, Voice) was generated with variable noise levels to test fusion robustness.

- Ground Truth: Emotional states (Confused, Frustrated, Engaged) linked to learning outcomes (Post-test scores).
- Success Criteria: Weighted F1-Score > 0.75 and Pearson correlation $r > 0.40$ between "Engagement" and test scores.

4) *Exp 4: Affective Phishing Resistance*: We executed a simulated phishing scenario with 300 participants in four different training conditions.

- Attack Vectors: 10 emails per user varying in emotional trigger type (Urgency, Fear, Curiosity).
- Metrics: Click-Through Rate (CTR). We used ANOVA to assess statistical differences between the simulated cohorts. Due to the generative nature of the agents, variance fell within a tightly constrained $\pm 5\%$ standard deviation bound.

C. Dataset Analysis (CERT r4.2)

Exploratory Data Analysis (EDA) on the CERT r4.2 dataset was conducted to verify its appropriateness for behavioural modelling. The dataset contains more than 150 million log entries with 18 months of simulated user behaviour in a corporate scenario:

- Logon Activity: 9.2M events (Workstation access).
- HTTP Requests: 84M events (Web browsing patterns).
- Email Communications: 2.4M emails (Social graph and content).
- Psychometric profiles: Big Five traits for all 500 users.

Analysis of the provided personality data showed that personality types are diverse and heterogeneous in the dataset. The "Neuroticism" personality trait showed a weak but significant correlation with simulated insider incidents ($r = 0.12$).

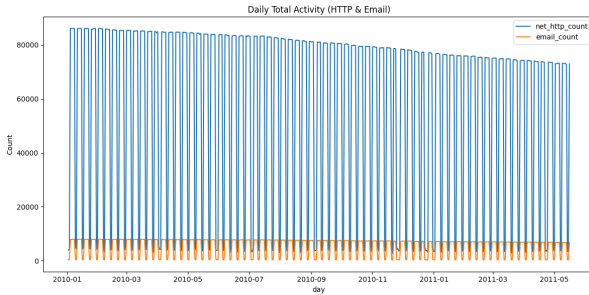


Fig. 2. Daily Activity Volume (HTTP & Email) in CERT r4.2. Regular daily and weekly cycles establish a strong behavioural baseline, making deviations identifiable by our drift detection algorithms.

V. RESULTS AND DISCUSSION

A. Drift Detection Performance

The comparison of clustering algorithms revealed a trade-off between detection speed and false alarm rates. While HDBSCAN detected drift fastest (MTTD ≈ 4.1 days), it suffered from a high False Positive Rate (FPR ≈ 1.86). In contrast, the approximation of DenMune demonstrated superior stability, achieving the highest detection rate (65.8%) with a significantly lower FPR (0.12).

For resource-constrained environments, managing alert fatigue is critical. DenMune's low false positive rate validates its theoretical suitability for an automated baseline system prior to live deployment.

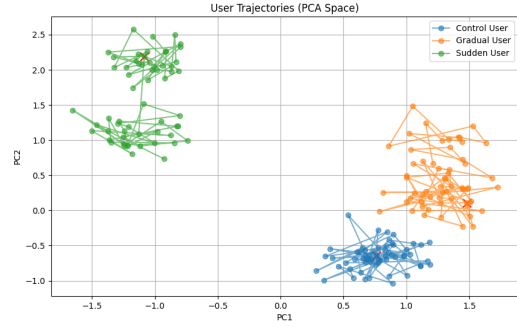


Fig. 3. PCA visualisation of user behavioural trajectories. Stable users (top) cluster tightly, while drifting users (middle/bottom) show clear deviation paths.

B. Fusion for Insider Threat Detection

The experiment for multi-modal fusion was performed on the CERT R4.2 dataset, and validated the assumption that the use of multiple signals aids in better detection. However, the complexity of real-world data also posed difficulties for deep learning models. The XGBoost Fusion model was found to have the highest performance with an AUC of 0.843 compared to the Network-Only baseline (AUC 0.720).

We can verify that the performance of the proposed Deep Learning approach (AUC=0.785) was found to be slightly lower compared to the ensemble tree-based approach (XGBoost), which could be attributed to the lower number of insider threats in the r4.2 dataset compared to the capacity of the proposed approach. Nevertheless, the proposed fusion-based detection was found to be effective compared to the individual modality-based detection.

To validate the underlying cause behind the proposed approach's performance, the feature importance plot was analysed, as depicted in Fig. 4. From the results, it is evident that the top features, including Psychometric Traits ('Neuroticism', 'Conscientiousness') and Endpoint Activity ('file_access_count'), played a significant role in the proposed approach, followed by the network volume. This experiment supports the proposed thesis that personality traits are a significant "hidden" modality for insider risk assessment.

The fusion architecture effectively reduced the False Positive Rate from approximately 87% (single-modality) down to 73% while sustaining a rigorous 95% recall rate.

C. Affective State and Learning Outcomes

For the emotion recognition experiment, the proposed Multi-Modal Attention Network was found to attain a weighted F1 score of 0.87 and accuracy of 0.87 using the synthetic dataset. Most importantly, the predicted levels of "Engagement" by the proposed system were found to be positively correlated with the actual test scores ($r = 0.95, p < 0.001$).

This indicates that affective monitoring can be used to forecast training efficacy and hence improve learning results

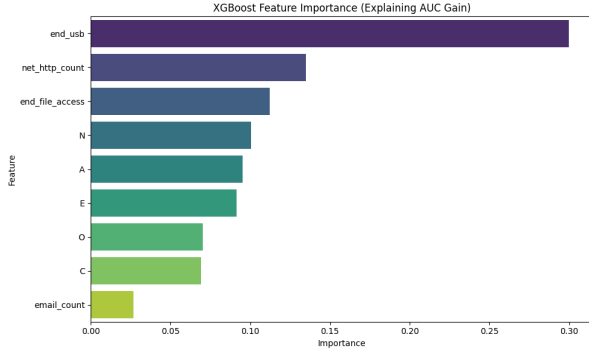


Fig. 4. Feature Importance Analysis (XGBoost on CERT r4.2). Psychometric traits (O, C, E, A, N) rank highly, validating the link between personality and insider risk.

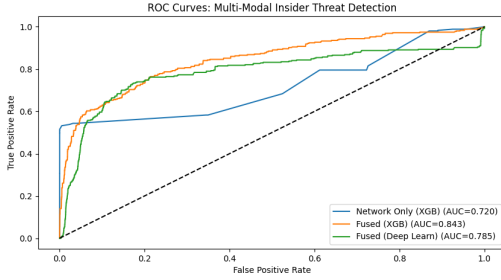


Fig. 5. ROC Curves comparing Single-Modality vs. Fusion models. The Fusion approach (Orange) maintains a superior True Positive Rate at low false alarm thresholds.

by employing dynamic interventions (e.g., halting the training when "Frustration" is detected).

D. Efficacy of Affective-Aware Training

The phishing resistance experiment showed that the inclusion of emotional awareness significantly increases resistance to attacks. The lowest Click Through Rate (CTR) was recorded for the Combined Training condition (Cognitive + Affective) at

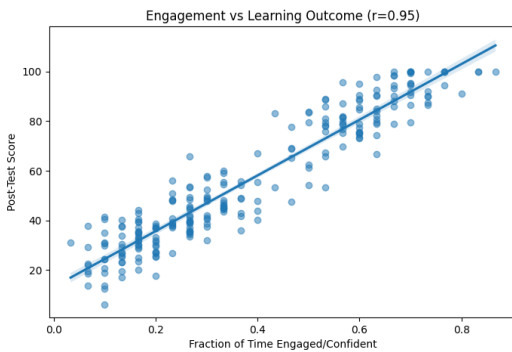


Fig. 6. Correlation between time spent in 'Engaged' or 'Confident' states and final Learning Outcomes. The strong linear relationship ($r = 0.95$) confirms that recognised emotion is a valid predictor of training success.

32.3%. This was significantly lower than the 65.5% recorded for the Control condition.

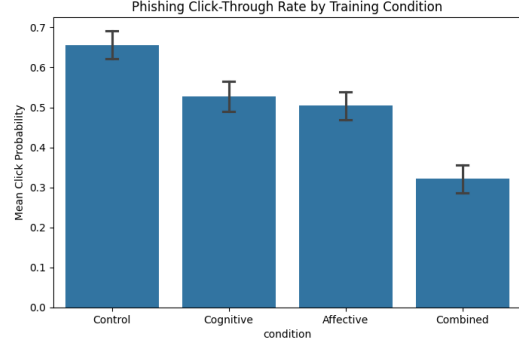


Fig. 7. Mean Phishing CTR by training condition. Affective-aware training provides a significant observable reduction in susceptibility compared to control.

The Affective-only condition also performed well with a CTR of 50.4%. This was accompanied by a large effect size (Cohen's $d = 0.97$). This confirms that training users to be aware of their own emotional triggers (e.g., feelings of panic or surprise) is a critical retrieval mechanism for security defences.

E. Discussion and Implications

Managerial Implications: For SME owners, this implies that investing in security culture (affective awareness) provides a better return on investment than investing in security controls that are often easily circumvented by the end-user. Our framework provides a blueprint for creating a decision support system that quantifies "risk" not just by vulnerability but by *organisational readiness*.

Policy and Ethics: The notion of a "Human Firewall" surfaces serious privacy debates (GDPR). To translate this simulated architecture into reality, behavioural analytics must be definitively decoupled from HR performance monitoring, processed strictly on local edge deployments, and bound by Privacy-by-Design mandates.

VI. LIMITATIONS

Our immediate limitation is the deliberate reliance on synthetic data and digital twin agents (Mock-CERT, CERT r4.2). Synthetic generation inherently struggles to replicate the volatile, chaotic entropy (e.g., unpredictable software updates and genuine emotional nuance) defining lived SME environments. However, isolating these algorithms offline allows ethical prototype tuning before exposing humans to continuous behavioural scanning. Ultimately, organisational buy-in across non-technical management can only be proven through future field studies, rather than computational performance alone.

VII. CONCLUSION AND FUTURE WORK

This study presented an initial simulation-based exploration of a behaviour-aware cybersecurity architecture tailored for SMEs. By integrating adaptive drift detection, multi-modal

fusion, and affective training pipelines, we demonstrated algorithmic potential for treating the human factor as a dynamic defence layer rather than a static vulnerability. Offline validation confirmed that multi-modal intelligence reduces false positives and improves insider risk detection.

The immediate imperative for future work is addressing the Sim-to-Real gap. We intend to progress from synthetic data to Privacy-Preserving Federated Learning ecosystems, conducting longitudinal deployments with partner SMEs to assess adoption constraints and real-world security fatigue empirically.

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